

Related Work and Research Gap: A Graph Neural Network Approach for EMG-Based Gesture Recognition on NinaPro DB1

Group 04

Related Work

Table 1: Summary of related work (2022–2026) on GNN/GCN-based physiological-signal gesture recognition.

Authors (Year)	Title	Dataset	Application Area	Method/ Model	Metrics	Strengths	Limitations / Gap	Relation to Our Work
Xu, Chen, Ruan, & Zhang [1] (2023)	"Research on electromyography pattern recognition based on graph neural network"	HD-sEMG (multi-subject)	Cross-user EMG pattern / gesture recognition	Spatio-temporal GCN (CNN-MSTGCN)	Accuracy, F1-score	Combines CNN feature extraction with multi-scale spatio-temporal graph module; targets cross-user generalization	Channel-graph topology is fixed rather than learned per gesture; HD-sEMG focus limits applicability to sparse electrode setups	Our work uses correlation-driven graph topology on low-channel NinaPro DB1, enabling adaptive graph learning rather than fixed physical adjacency
Lee et al. [2] (2023)	"Stretchable array electromyography sensor with graph neural network for static and dynamic gestures recognition system"	Custom stretchable 18-gesture array	Static/dynamic hand gesture recognition	GNN on flexible-array electrodes	~97% accuracy	Pairs stretchable sensor array with spatially-aware graph model; high accuracy on dynamic gestures	Custom hardware/dataset not compatible with public EMG benchmarks; limited reproducibility	We use public NinaPro DB1 benchmark to ensure reproducibility and comparability with existing methods
Massa, Riboni, & Nazarpour [3] (2023)	"Explainable AI-powered graph neural networks for HD EMG-based gesture intention recognition"	HD-sEMG, 65 gestures, 20 subjects	Fine-grained gesture-intention recognition	GNN + GNNExplainer topology refinement (xAI-GNN)	~8.75% average error	Uses explainable AI to prune uninformative edges, improving interpretability and efficiency	Preliminary version reported high variance across gestures; HD-sEMG only; no class imbalance handling	Our approach addresses class imbalance explicitly in training strategy, applicable to standard 10-electrode configuration

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Lee, Jiang, Yang, Yang, & Zhao [4] (2025)	"Decoding gestures in electromyography: Spatiotemporal graph neural networks for generalizable and interpretable classification"	Five public EMG datasets	Generalizable, interpretable gesture classification	Spatiotemporal GCN	State-of-the-art accuracy across datasets	Graph encodes both spatial electrode proximity and temporal adjacency; strong cross-dataset performance	Prioritizes cross-dataset generalization over single-subject, low-channel deployment; fixed graph topology	Our work focuses specifically on single-subject, low-channel NinaPro DB1 with correlation-driven graph construction
Massa, Riboni, & Nazarpour [5] (2022)	"Graph neural networks for HD EMG-based movement intention recognition: An initial investigation"	HD-sEMG, 65 gestures, 20 subjects	Movement-intention recognition (initial study)	Baseline GNN on HD-EMG channel graph	Preliminary accuracy metrics	Early proof-of-concept that GNNs are viable for raw-EMG-channel gesture decoding	Simple, non-adaptive graph topology; only two HD-EMG arrays tested	We extend from HD-sEMG to low-channel sEMG and use correlation-based adaptive graph topology
Vipulanandan, Premaratne, & Murthi [6] (2024)	"A graph neural network model for real-time gesture recognition based on sEMG signals"	Myoband sEMG, 8 electrodes, 8 subjects	Real-time hand gesture recognition	GNN on correlation-based signal graph	99% accuracy	Graph built directly from inter-channel signal correlation, not fixed layout	Evaluated on only 5 gesture classes and a small subject pool; limited generalization assessment	We extend correlation-based graph construction to 10 channels and 12 gesture classes with rigorous subject-independent evaluation
Li, Ji, Yu, Li, Jin, Liu, Bai, & Ye [7] (2023)	"A sequential learning model with GNN for EEG-EMG-based stroke rehabilitation BCI"	EEG + EMG, stroke rehabilitation cohort	Multimodal stroke rehabilitation BCI	GIN (Graph Isomorphism Network) on heterogeneous graph	Classification accuracy, AUC	Combines EEG and EMG as heterogeneous graph using mutual information for edges; strong gains over baselines	Heterogeneous EEG-EMG graph adds complexity not needed for EMG-only setups; stroke-specific context	We focus on EMG-only graph construction for gesture recognition, avoiding unnecessary multimodal complexity

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Quadrelli, Canepa, Di Domenico, Boccardo, Chiappalone, & Lafranchi [8] (2025)	"Advances in HD-EMG interfaces and spatial algorithms for upper limb prosthetic control"	HD-EMG interfaces (review scope)	Upper-limb myoelectric prosthetic control	Review of GNN and spatial-ML approaches	N/A (it is a review paper)	Surveys spatial/graph-based ML exploiting HD-EMG electrode layout to improve intention detection	Mini-review scope; does not benchmark a single model on standard low-channel dataset	Provides context for graph-based EMG research but highlights gap in low-channel sEMG GNN applications
Klepl, He, Wu, Blackburn, & Sarrigiannis [9] (2024)	"Graph neural network-based EEG classification: A survey"	Cross-dataset (EEG classification survey)	Systematic review of GNN-based physiological classification	Survey of spectral and spatial GNN variants	N/A (it is a review paper)	Identifies that spectral graph convolutions dominate current GNN-based biosignal classifiers	Focused on EEG; does not cover low-channel sEMG graph construction specifically	Emphasizes the need for EMG-specific GNN studies; our work addresses this gap for sEMG
Hou, Jia, Lun, Hao, Shi, Li, Zeng, & Lv [10] (2022)	"GCNs-Net: A graph convolutional neural network approach for decoding time-resolved EEG motor imagery signals"	EEG motor-imagery benchmark datasets	Time-resolved motor-imagery decoding	GCNs-Net (graph convolutions + pooling)	Accuracy, Kappa score	Learns generalized features respecting electrode topology; robust across individual variability	Designed for EEG electrode topology; not validated on sparse EMG channel graphs	We adapt graph convolution principles from EEG to EMG, with topology informed by signal correlation rather than physical layout

Research Gap

Identified Gaps in Current Literature

1. **HD-sEMG Concentration:** Existing GNN/GCN work on EMG gesture recognition is concentrated on high-density (HD) sEMG setups with dozens to over a hundred electrodes [1, 3, 4]; standard low-channel (10-electrode) surface EMG benchmarks such as NinaPro DB1 remain largely unexamined with GNNs.

2. **Fixed vs. Learned Topology:** Graph topology in most prior work is fixed by physical electrode adjacency rather than learned from inter-channel signal correlation, which is the more relevant cue when only 10 sparsely-placed channels are available [5].
3. **Hardware-Specific Studies:** Custom-hardware studies [2] report strong accuracy but on non-public, non-comparable datasets, limiting reproducibility against standard EMG benchmarks.
4. **EEG-Dominated Research:** Most GNN-based bio-signal survey and review work to date centers on EEG [9, 10], with comparatively few systematic studies applying the same graph-construction principles to low-channel sEMG.
5. **Class Imbalance Neglect:** None of the reviewed studies explicitly address severe rest-vs-gesture class imbalance within the graph construction or training strategy itself, despite this being a dominant characteristic of raw EMG streams.

Our Contribution

This work closes these gaps by:

- Building a **correlation-driven channel graph** from a standard low-density NinaPro configuration
- Evaluating under a **subject-independent, leakage-free split** with explicit **class-imbalance handling**
- Providing a **reproducible benchmark** on public NinaPro DB1 data
- Extending graph-based EMG research from HD-sEMG to **low-channel sEMG** applications

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